Logic Gates, Multiplexers and Decoders

1. Logic Gates

Logic gates are basic digital electronic circuits that perform logical operations on binary inputs (0 and 1) to produce a binary output.

Types of Logic Gates

Truth Tables

A	B	AND (A·B)	OR (A+B)	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

NOT Gate

A	Output ($\overline{A}$)
0	1
1	0

Boolean Expressions

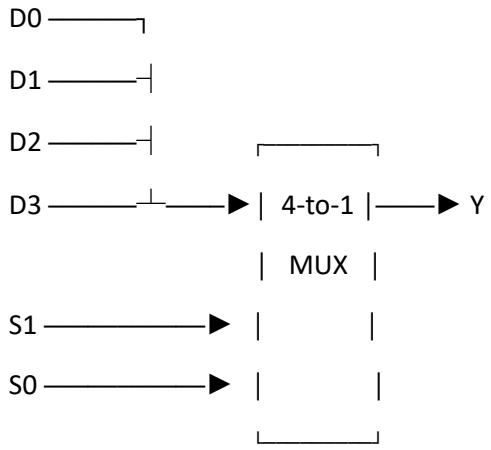
- AND: $Y = A \cdot B$
- OR: $Y = A + B$
- NOT: $Y = \overline{A}$
- NAND: $Y = \overline{(A \cdot B)}$
- NOR: $Y = \overline{(A + B)}$
- XOR: $Y = A \oplus B$
- XNOR: $Y = \overline{(A \oplus B)}$

NAND and NOR are called universal gates because any basic logic circuit can be constructed using either one alone.

2. Multiplexer (MUX)

A multiplexer is a combinational circuit that selects one input from several input lines and sends it to a single output line. Selection is controlled by select lines.

4-to-1 Multiplexer Diagram



Selection Table

S1	S0	Output Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Boolean expression:

$Y = D0 \cdot \overline{S}1 \cdot \overline{S}0 + D1 \cdot \overline{S}1 \cdot S0 + D2 \cdot S1 \cdot \overline{S}0 + D3 \cdot S1 \cdot S0$

Important Points

- A 2-to-1 MUX has 2 inputs and 1 select line.
- A 4-to-1 MUX has 4 inputs and 2 select lines.
- An 8-to-1 MUX has 8 inputs and 3 select lines.
- For \(2^n\) inputs, the number of select lines is n.

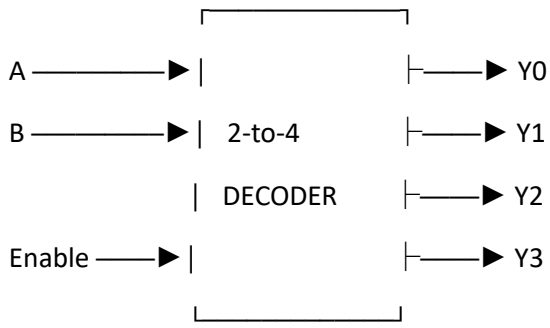
Applications

- Data selection and routing
- Communication systems
- Processor datapaths
- Digital signal processing

3. Decoder

A decoder is a combinational circuit that converts an n-bit binary input into a maximum of \(2^n\) distinct output lines. For each valid input combination, the corresponding output is activated.

2-to-4 Decoder Diagram



Truth Table

Assume the decoder is enabled and has active-high outputs.

A	B	Y0	Y1	Y2	Y3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Boolean Expressions

- $Y0 = \overline{A} \cdot \overline{B}$
- $Y1 = \overline{A} \cdot B$
- $Y2 = A \cdot \overline{B}$
- $Y3 = A \cdot B$

Applications

- Memory address decoding
- Instruction decoding in processors
- Display systems
- Digital control circuits

4. Difference Between MUX and Decoder

Multiplexer	Decoder
Selects one input and forwards it to one output.	Activates a specific output based on a binary input.
Has select lines.	Has binary input lines and may have an enable input.
Commonly used for data routing.	Commonly used for address and instruction decoding.

5. Result

Logic gates perform basic Boolean operations. A multiplexer selects one of several inputs using select lines, whereas a decoder activates a specific output corresponding to a binary input combination.